FLUENT WORKFLOW - NOSE CONE CFD

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Weisshorn – Nose Cone CFD Workflow

EPFL ROCKET TEAM

Weisshorn nose cone: Von Kármán, $L = 810\,\text{mm}$, $D = 180\,\text{mm}$, tip radius 4.5 mm. Verification runs of recommendation R6 of 2026_C_FD_REPORT-1_NOSE_CONE.

1 Solver: Fluent

Use **Fluent** (the "Fluid Flow (Fluent)" system in Workbench) with these settings: density-based, implicit, steady, k-ω SST, energy on, air as an ideal gas with Sutherland viscosity. That is the setup the nose cone report already prescribes (R6 and Section 8), and it is what the team runs on the DLL VM (Fluent 2024 R2). Discovery is fine as the geometry editor inside Workbench, but do not use its own fluid simulation for shocks.

Two models, not one.

1. **2D axisymmetric** for the three zero-incidence runs ($M = 0.95$, 1.2, 1.47). It is cheap, it resolves the boundary layer and the shocks properly, and it compares directly with the axisymmetric Euler results.
2. **3D half-model** with a symmetry plane, used only for the 5° incidence run ($C_{N\alpha}$ and CP). The STEP file goes here.

Input files.

- Design solid (STEP): `nose_vonkarman_LD4.5_rn4.5mm_FINAL.step`, in `output/weisshorn/cad/` of this repository. `output/` is not tracked by git; `python -m nosecone export configs/weisshorn.yaml` writes it.
  Use the STEP, not the STL: the STL opens as a faceted body (about 122 600 triangles), on which subtracting it from a flow enclosure, naming faces and meshing do not work cleanly. The STEP is a true solid of revolution (nose, 60 mm shoulder, base disc) in mm, with the x axis as the rocket axis and the tip at the origin. It is a filled solid with no wall thickness, which is what external-flow CFD needs.
- Profile points: `nose_profile_final.csv` (columns `x_mm`, `r_mm`), same folder.
- Euler reference, in `data/`: `euler_results.csv` and `cfd/cp/f45_vonkarman_f4.5_rn0.05_medium_M*.npz` ($M = 1.2$, 1.4, 1.6).

Free-stream conditions from the nominal trajectory:

M	altitude	V	p	T	Re_L	first cell for $y^+ \approx 1$
0.95	928 m	321 m/s	90 666 Pa	282.1 K	$1.7 \cdot 10^7$	1.4 µm
1.20	1478 m	402 m/s	84 789 Pa	278.5 K	$2.0 \cdot 10^7$	1.2 µm
1.47	2189 m	488 m/s	77 649 Pa	273.9 K	$2.3 \cdot 10^7$	1.0 µm

The first-cell heights are flat-plate estimates, so check y^+ after the first run. The hot-day case reaches $M = 1.56$ (at 2315 m) if you want one extra run.

2 2D workflow

1. **Workbench:** add a “Fluid Flow (Fluent)” system. **Before you open the geometry**, set Geometry cell $\rightarrow$ Properties $\rightarrow$ Analysis Type = **2D**. If you skip this, Fluent will not let you choose axisymmetric.
2. **Geometry:**
   - Build one surface in the XY plane with $y \geq 0$: a far-field rectangle minus the nose half-section, with the x axis as the rocket axis. The STEP already uses x as the axis, and `nose_profile_final.csv` gives the profile points.
   - Run the cylinder straight through to the outlet. With no base there is no wake, so you get clean forebody numbers, the same as the Euler nose-plus-cylinder setup.
   - Domain size: about $20D$ upstream and radially (3.6 m) and about $10D$ behind the shoulder.
   - Split the wall at the shoulder and add named selections: `inlet`, `farfield`, `outlet`, `axis`, `wall_nose`, `wall_body`.
3. **Mesh** (Ansys Meshing):
   - Inflation on both walls: 1 µm first layer, growth 1.2, about 45 layers. That is about 18 mm in total, which covers the roughly 10 mm boundary layer at the shoulder.
   - Refine the tip to about 0.2 mm cells, since the tip radius is only 4.5 mm.
4. **Fluent setup:**
   - Launcher: 2D, double precision. General: Density-Based, Steady, Axisymmetric.
   - **Operating pressure = 0 Pa**, so every pressure you enter is absolute.
   - Boundary conditions:
     - `inlet` and `farfield`: pressure-far-field with M, p and T from the table, flow direction $(1, 0)$.
     - `outlet`: pressure-outlet at p_∞.
     - `axis`: axis.
     - Walls: no-slip, adiabatic.
   - Reference values: area $\pi \cdot 0.18^2/4 = 0.025\,45\ \text{m}^2$, length 0.18 m. Fluent integrates axisymmetric forces over the full 360°, so this gives the same C_D definition as the Euler run.
   - Methods: Roe-FDS flux, least-squares gradients. Run first order for about 300–500 iterations, then switch to second order. Start the Courant number at 1 and raise it to 5–10 once the run is stable.
   - Initialise with FMG or hybrid initialisation, starting from the far-field values.
5. **Convergence:** monitor the drag coefficient on `wall_nose`. Fluent splits it into pressure and viscous parts. The run is converged when C_D changes by less than 0.1 % over about 500 iterations. Residuals alone do not tell you that.
6. **Mesh study:** at $M = 1.47$ only, run coarse, medium and fine meshes (about $\times 1.5$ per direction each step), plus one or two pressure-gradient adaptions at the shock.
7. **Compare with the Euler results:**
   - Fluent’s **pressure** part should match $C_{D,\text{w}}$. $M = 1.2$ is directly in `euler_results.csv`; $M = 1.47$ falls between the 1.4 and 1.6 values.
   - Plot C_p along the nose against `data/cfd/cp/f45_vonkarman_f4.5_rn0.05_medium_M*.npz`.
   - $M = 0.95$ replaces the cubic transonic model of the report.

3 3D run at 5° incidence

- Use a second Fluent system in 3D with a half domain cut on the XY plane, and set that plane as a symmetry boundary. Flow direction is $(\cos 5°, \sin 5°, 0)$.
- For the mesh, Fluent Meshing (watertight workflow, poly-hexcore with prism layers) handles this better than Ansys Meshing.
- Normal force is dominated by pressure, so $y^+ \approx 30$–50 (first cell about 30 µm to 50 µm) is acceptable here and keeps the cell count down.
- Report the normal force and the moment about the tip.
  - A half model gives half the force, so double it (or halve the reference area).
  - $C_{N\alpha} = C_N/0.0873$; compare it with 2 rad^{-1}.
  - $x_{\mathrm{CP}} = M_{\mathrm{tip}}/N$; compare it with 405 mm.

4 Running Fluent from Claude Code on the DLL VM

Claude Code cannot be installed onto the VM from the laptop, even while you are logged in: the Remote Desktop window gives no shell on the VM. Run the install yourself in PowerShell on the VM. It takes about 10 minutes following 2026_C_FD_REPORT-2_MCP-FLUENT:

- **Section 3.4** to install and sign in (`claude`), make the `.claude` junction and stage the copy on `Z:\`.
- **Sections 3.5–3.6** for the wheel cache and the repo clone. You need the repo URL from the guide's author, because the guide only has `<project-git-url>`.
- **Section 4.2** to install PyFluent and the MCP package.
- **Section 6.2** to register the MCP server, then **Section 7.1** to connect to a running Fluent.

Things to know.

- `C:\` **may be wiped** between sessions, so redo Section 6.2 when that happens.
- Write `.mcp.json` **without a BOM**.
- Start the server with `server/start-server`, not the Web Server menu.
- The copy on the VM is a separate installation. It starts without the context or the project notes of the Claude Code sessions on the laptop.
- The MCP server only drives the Fluent solver: reading a case, initialising, iterating, TUI and Scheme commands, screenshots. Geometry and meshing stay manual. It is useful for the Mach sweep and the mesh study, by changing the far-field values through TUI commands and re-running.